

Branch: B. Tech

Semester: III

Subject Name: Electronic devices and Circuits

Subject Code: BTEXC303

Max Marks: 60

Date:- 05/12/2018

Duration:- 3 Hrs.

Instructions to the Students:

1. Each Question Carries 12 Marks
2. Attempt any five questions of the following
3. Illustrate your answers with neat sketches, diagram etc., wherever necessary
4. If some part or parameter is noticed to be missing, you may appropriately assume it and should mention it clearly

Q.1 Attempt Any Two questions of the following

- | | |
|---|----|
| a) Briefly Explain construction and working principle of JFET | 06 |
| b) Describe FET as an Amplifier and its Analysis(CS) | 06 |
| c) Explain about FET as High Impedance Circuits | 06 |

- | | |
|---|----|
| Q.2 a) Briefly Explain the Characteristics of N-Channel E-MOSFET | 06 |
| b) Write short Notes on Non-Ideal Voltage current characteristics | 06 |

- | | |
|--|----|
| Q.3 a) Explain the Classification of amplifier in details | 06 |
| b) Discuss in details about various feedback topologies | 06 |

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|---|----|
| Q.4 Attempt Any Two questions of the following | |
| a) Explain about Barkhausen criterion for oscillator | 06 |
| b) Draw and explain the RC Phase shift Oscillators | 06 |
| c) Describe the Hartley and colpitts oscillators | 06 |

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|--|----|
| Q.5 a) Briefly Explain about the IC 555 Block Diagram | 06 |
| b) Write down the applications of IC 555 in Engineering | 06 |

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|---|----|
| Q.6 Attempt Any Two questions of the following | |
| a) Explain about adjustable three terminal positive (LM 317) regulators with neat diagram | 06 |
| b) Briefly Explain SMPS with Neat diagram | 06 |
| c) Write Down the Comparison of Linear power supply and SMPS | 06 |

*** End ***

Dr. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE-402103

Department of Electronics and Telecommunication Engineering

Supplementary Examination (MAY 2019)

Subject: Electronic Devices and Circuits (BTEXC303)

Class: B. Tech

Semester: III

Date: 30/ 05 /2019

Time: 3 h

Max. Marks: 60

Instructions to the candidates:

1. Question 1 is compulsory, attempt any five from question 2 to question 7
 2. Neat diagrams and waveforms must be drawn wherever necessary.
 3. Sub question from each main question must be solved together
 4. Use of nonprogrammable calculator is allowed.
 5. Assume suitable data if necessary.
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Q 1. Answer the following (2 Marks each) 10 M

- a. Given $I_{DSS} = 12 \text{ mA}$ and $V_P = -4 \text{ V}$, sketch the transfer characteristics for the JFET resistor.
- b. Define Transconductance g_m
- c. For a JFET, the typical values of amplification factor & transconductance are given as 40 & 100 μS respectively. The dynamic drain resistance of JFET will be?
- d. Draw the pin diagram of IC 555.
- e. Define barkhausen criterion.

Q 2. a. Draw and explain drain characteristics of n channel JFET 5 M

- b. Compare Common source, common drain, and common gate configuration of JFET 5 M**

Q 3. a. Datasheet for n channel EMOSFET specifies following parameter, $V_{GS}=10\text{V}$, $I_{D(on)}=500\text{mA}$, if $V_{GS(th)}$ for MOSFET is 1 V, determine the drain current for $V_{GS}=4\text{V}$? 5 M

- b. Draw and explain CMOS inverter characteristics 5 M**

Q 4. a. Compare class A, class B, class AB, class C and class D amplifier 5 M

- b. Compare different types of feedback topology based on voltage gain, bandwidth, input resistance & output resistance. 5 M**

- Q 5. a. With neat diagram explain hartley oscillator and derive an expression for frequency of oscillation. 5 M
- b. An FET phase-shift oscillator having $g_m = 6000 \text{ mS}$, $r_d = 36 \text{ k}\Omega$, and feedback resistor $R = 12 \text{ k}\Omega$ is to operate at 2.5 kHz . Select C for specified oscillator operation. 5 M
- Q 6. a. Draw and explain internal block diagram of IC 555. 5 M
- b. With a neat diagram, explain bistable multivibrator using IC 555. 5 M
- Q 7. a. Differentiate SMPS with linear regulated power supply. 4M
- b. Find the % load regulation of a power supply providing 100V unloaded and 95V at full load. 3 M
- c. Explain the need of regulated power supply. 3 M

*****BEST OF LUCK*****

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY,
LONERE – RAIGAD -402 103**

Winter Semester Examination – December - 2019

Branch: Electronics and Telecommunication Engineering

Sem.:- III

Subject :- Electronic Devices & Circuits (BTEXC303)

Marks: 60

Date:- 14/12/2019

Time:- 3 Hr.

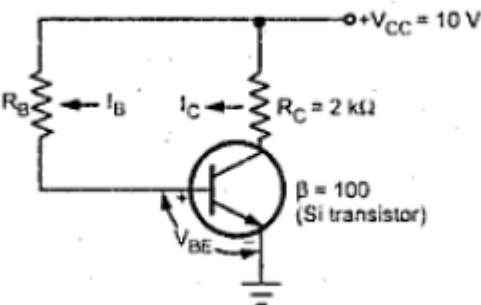
Instructions to the Students

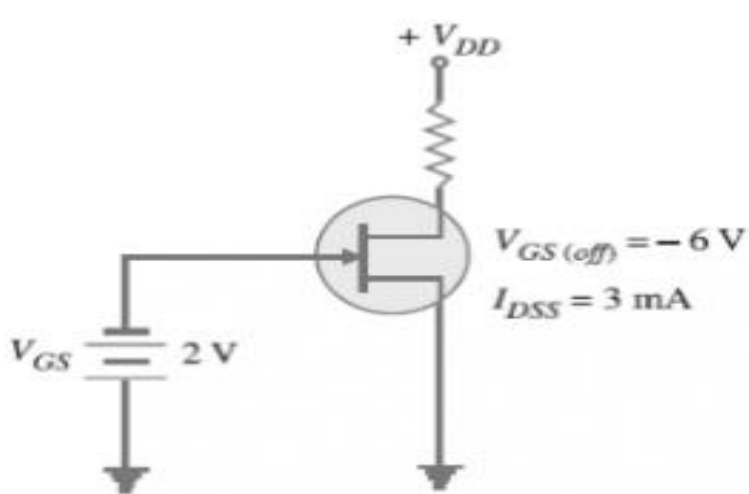
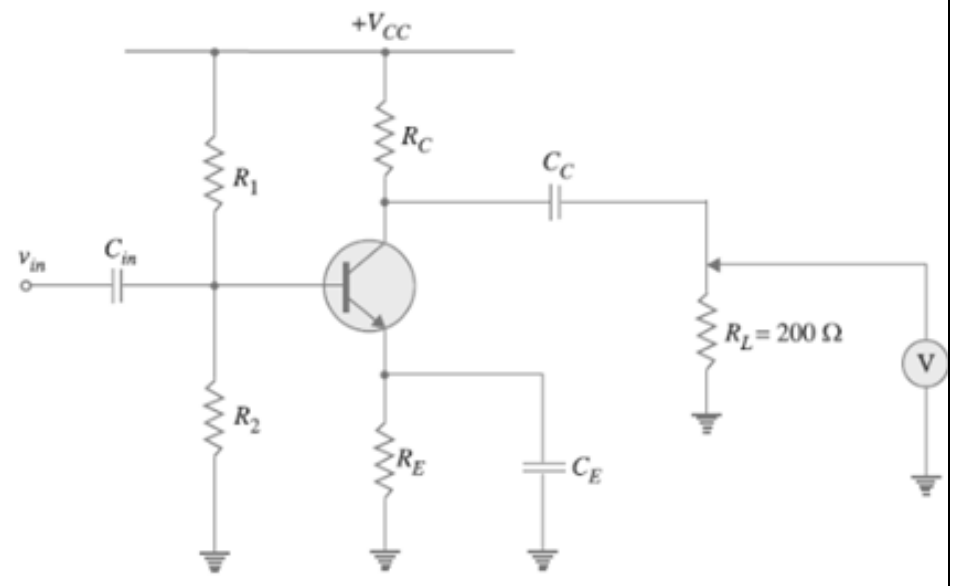
1. Each question carries 12 marks.
2. Attempt **any five** questions of the following.
3. Illustrate your answers with neat sketches, diagram etc., wherever necessary.
4. If some part or parameter is noticed to be missing, you may appropriately Assume it and should mention it clearly

	(Marks)
Q.1.A) Draw and explain contraction of n channel JFET and Compare Common source, common drain, and common gate configuration of JFET.	6
B) Explain the VI characteristics of JFET. What factors are responsible for the shape of the characteristics in a different region?	6
Q.2.A) Explain n channel enhancement MOSFET and Datasheet for n channel EMOSFET specifies the following parameter, $V_{GS}=10V$, $I_{D(on)}=500mA$, if $V_{GS(th)}$ for MOSFET is 1 V, determine the drain current for $V_{GS}=4V$?	6
B) Explain CMOS inverter with circuits. And draw & explain characteristics of CMOS inverter.	6
Q.3.A) Compare class A, class B, class AB, class C and class D amplifier.	6
B) Draw the circuits of voltage series feedback amplifier and derive the expressions for input impedance R_{if} .	6
Q.4.A) With a neat diagram, explain hardy oscillator and derive an expression for the frequency of oscillation. Find the frequency of oscillation if $L1=L2=10\text{ mH}$ and $C=0.1\text{ }\mu F$.	6
B) With a neat diagram, explain the RC phase shift oscillator and derive an expression for the frequency of oscillation. And Calculate the value of $C1 = C2$ for the Wien bridge oscillator to operate at a frequency of 20 kHz. Assume $R1 = R2 = 50\text{ k}$ and $R3 = 3, R4 = 600?$	6
Q.5.A) Draw and explain internal block diagram of IC 555 and explain the working of Astable multivibrator	6
B) Draw and explain a Monostable and bistable multivibrator.	6
Q.6.A) Differentiate SMPS with linear regulated power supply	4
B) Find the % load regulation of a power supply providing 100V unloaded and 95V at full load.	4
C) Draw and explain the working principle of IC LM317	4

	DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE			
	Supplementary Examination – Summer 2022			
	Course: B. Tech.	Branch: E&TC	Semester: III	
	Subject Code & Name: BTETC302 Electronic Devices and Circuits			
	Max Marks: 60	Date:	Duration: 3 Hr.	
	Instructions to the Students: 1. All the questions are compulsory. 2. Use of non-programmable scientific calculators is allowed. 3. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks	
Q. 1	Solve Any Two of the following.		12	
A)	Draw and explain RC coupled CE Amplifier. What are Effects of bypass and coupling capacitors.	(1)	6	
B)	What is Need for biasing BJT also explain Transistor biasing methods.	(2)	6	
C)	Explain BJT: construction, working, characteristics.	(1)	6	
Q.2	Solve Any Two of the following.		12	
A)	Briefly Explain construction and working principle of D-MOSFET	(2)	6	
B)	Describe FET as an Amplifier and its Analysis (CS).	(3)	6	
C)	Compare JFET, MOSFET and BJT w.r.t. to device and Circuit parameter	(2,3)	6	
Q. 3	Solve Any One of the following.		12	
A)	What is Class B push pull and complementary symmetry amplifier with its efficiency and calculate the power output and power dissipation.	(3)	12	
B)	Compare Class A, Class B, Class C, Class AB and Class D amplifiers.	(4)	12	
Q.4	Solve Any Two of the following.		12	
A)	Briefly Explain Principle of Negative feedback in electronic circuits.	(4)	6	
B)	Explain Voltage series, Voltage shunt, Current series, Current shunt types of Negative feedback	(2)	6	
C)	What are the typical transistor circuits effects of Negative feedback on Input and Output impedance, Voltage and Current gains, and Bandwidth?	(3)	6	
Q. 5	Solve Any Two of the following.		12	
A)	Explain about Barkhausen criterion for oscillator also draw RC Phase shift Oscillators	(4)	6	
B)	Describe the Hartley and Colpitts oscillators.	(4)	6	
C)	Explain about adjustable three terminal positive (LM 317) regulators with neat diagram	(4)	6	
	*** End ***			

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DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Winter Examination – 2023 Course: B. Tech. Branch: E&TC Semester: III Subject Code & Name BTEXC302& Electronic Devices & Circuits Max Marks: 60 Date:11/03/2023 Duration: 2:00 To 5:00 PM			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		Level/(CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	Derive the relation between α & β with respect to BJT.	C01	6
B)	In the circuit shown below, for $R_B=300K\Omega$ and $R_C=150K\Omega$ calculate I_B, I_C and V_{CE}? 	C03	6
C)	Draw neat diagram of RC Coupled amplifier and explain in detail.	C03	6
Q.2	Solve Any Two of the following.		12
A)	Explain E-MOSFET in detail.	C03	6
B)	What is CMOS inverter? Explain its transfer characteristics.	C01	6
C)	Determine the value of drain current for the circuit shown in Fig.	C03	6

			
Q. 3	Solve Any Two of the following.		12
A)	Explain class c power amplifier in detail.	C02	6
B)	Determine the a.c. load power for the circuit shown in fig	C03	6
			
C)	Derive Expression for Maximum Efficiency of Class B Power Amplifier.	C02	6
Q.4	Solve Any Two of the following.		12
A)	What are the four different types of feedback amplifier? Explain in detail.	C04	6
B)	The overall gain of a multistage amplifier is 140. When negative voltage feedback is applied, the gain is reduced to 17.5. Find the fraction of the output that is feedback to the input.	C04	6
C)	Explain the principle of positive feedback.	C04	6

Q. 5	Solve Any Two of the following.		12
A)	Draw a neat diagram of clapp oscillator & explain in detail.	C01	6
B)	1 mH inductor is available. Choose the capacitor values in a Colpitts oscillator so that $f = 1$ MHz and feedback factor equal to 0.25	C04	6
C)	i)State Barkhausen criterion for sustained oscillation. ii)Differentiate oscillator from amplifier.	C04	6
	*** End ***		

The grid and the borders of the table will be hidden before final printing.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

Level/(CO) Marks

Q. 1) Solve Any Two of the following.

12

A) What is BJT? Explain in detail.

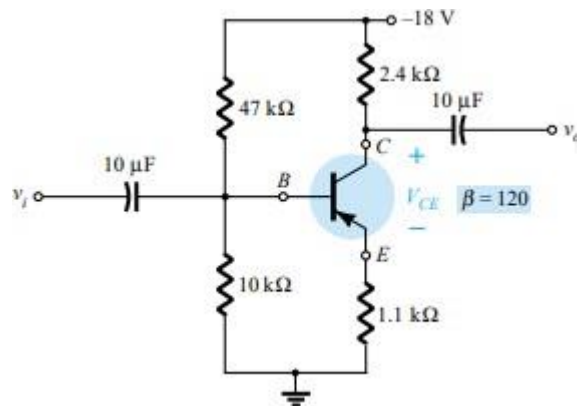
1/7

6

B) Determine V_{CE} for the voltage divider bias configuration?

1/3

6



C) Draw a neat diagram of cascaded amplifier and explain in detail.

1/3

6

Q. 2) Solve Any Two of the following.

12

A) Explain construction & characteristics of JFET.

2/1

6

B) Determine the following parameter for given figure

2/1

6

(a) V_{GSQ} .

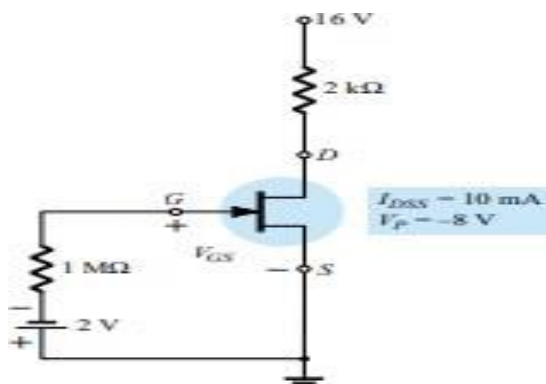
(b) I_{DQ} .

(c) V_{DS} .

(d) V_D .

(e) V_G .

(f) V_S .



C) Write short notes on CMOS.	1/5	6
Q. 3) Solve Any Two of the following.		12
A) Write short note on Transformer coupled class A power amplifier.	1/5	6
B) Derive Expression for Maximum Efficiency of Class B Power Amplifier?	2/1	6
C) Calculate the efficiency of a class B amplifier for a supply voltage of $V_{CC} = 24\text{ V}$ with peak output voltages of: (a) $V_L(p) = 22\text{ V}$. (b) $V_L(p) = 6\text{ V}$.	1/1	6
Q. 4) Solve Any Two of the following.		12
A) What is feedback? Explain its types in detail.	2/1	6
B) Determine the voltage gain, input, and output impedance with feedback for voltage series feedback having $A = -100$, $R_i = 10\text{ k}\Omega$, $R_o = 20\text{ k}\Omega$ for feedback of (a) $\beta = -0.1$ and (b) $\beta = -0.5$	1/7	6
C) Explain feedback amplifier in detail.	2/3	6
Q. 5) Solve Any Two of the following.		12
A) Explain RC phase shift oscillator in detail.	1/1	6
B) The tuned collector oscillator circuit used in the local oscillator of a radio receiver makes use of an LC tuned circuit with $L_1 = 58.6\text{ }\mu\text{H}$ and $C_1 = 300\text{ pF}$. Calculate the frequency of oscillations.	2/5	6
C) Write short note on Colpitts oscillator.	2/5	6

*** End ***

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Winter Examination-2023

Course: B. Tech.

Semester :III

Branch :Electronics Engineering/Electronics and Telecommunication

Engineering/Electronics and Communication Engineering

Subject Code & Name: BTEXC302/BTETC302 and Electronic Devices and Circuits

Max Marks: 60

Date:04-01-2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Draw a neat circuit diagram of CB connection. Define α . Find the expression for collector current.	CO3	6
B) Explain transistor as an amplifier in CE configuration.	CO3	6
C) Determine V_{CB} in the transistor circuit shown in figure (a). The transistor is silicon and has $\beta = 150$.	CO2/CO3	6

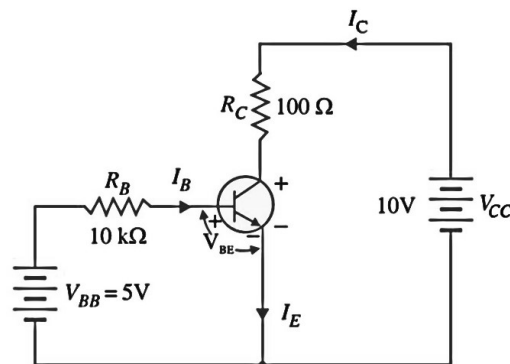


Figure (a)

Q.2 Solve Any Two of the following.		12
A) With a neat diagram, explain the working principle of p-channel D-MOSFET with different cases.	CO2/CO3	6
B) Give the comparison of FET with MOSFET and BJT.	CO3	6
C) A JFET has a drain current of 5mA. If $I_{DSS} = 10\text{mA}$ and $V_{GS(\text{off})} = -6\text{V}$, find the value of (i) V_{GS} and (ii) V_P .	CO3	6

Q. 3 Solve Any Two of the following.		12
A) Discuss with neat diagram transformer coupled class A power amplifier and derive its maximum efficiency.	CO2/CO4	6
B) Prove that the maximum conversion efficiency of a Class B transformer coupled amplifier is 78.5%.	CO2/CO4	6
C) With a neat diagram explain push-pull class B power amplifier and derive its maximum efficiency.	CO1/CO4	6
Q.4 Solve Any Two of the following.		12
A) Explain voltage series feedback amplifier with a neat diagram. How the overall gain, input impedance, output impedance is affected in these amplifiers.	CO4	6
B) Explain current series feedback amplifier with a neat diagram. How the input impedance, output impedance are affected in these amplifiers.	CO2/CO4	6
C) State and explain Barkhausen criterion of oscillations.	CO4	6
Q. 5 Solve Any Two of the following.		12
A) Draw and explain the Wein Bridge oscillator. Derive the expression for frequency of oscillation.	CO2/CO4	6
B) Draw and explain the Hartley oscillator. Derive the expression for frequency of oscillation.	CO2/CO4	6
C) Define voltage regulation. Explain series voltage regulator with a block diagram.	CO4	6

*** End ***

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Examination Summer 2024 Course: B. Tech. Semester : III Branch : Electronics Engineering / Electronics and Telecommunication Engineering Subject Code & Name: BTEXC303_Y18 Electronic Devices & Circuits Max Marks: 60 Date: 04/07/2024 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	Explain construction and working principle of JFET.	1	6
B)	Describe FET as an Amplifier and its analysis (CS)	2	6
C)	Explain FET used for High Impedance Circuit.	4	6
Q.2	Solve Any Two of the following.		12
A)	Draw and explain the biasing circuits for MOSFET using common source configuration.	1	6
B)	Briefly Explain the Characteristics of N-Channel E-MOSFET.	2	6
C)	Write short Notes on Non-ideal voltage current characteristics	3	6
Q. 3	Solve Any Two of the following.		12
A)	Compare Class A, Class B, class AB, class C and class D amplifier	1	6
B)	Draw the circuits of voltage series feedback amplifier and derive the expressions for input impedance R_{if} .	3	6
C)	Discuss in details about various feedback topologies.	2	6
Q.4	Solve Any Two of the following.		12
A)	Explain about Barkhausen criterion for oscillator.	1	6
B)	Describe the Hartley and Colpitts oscillators.	2	6
C)	An FET phase-shift oscillator having $g_m = 6000 \text{ mS}$, $r_d = 36 \text{ k}\Omega$, and feedback resistor $R = 12 \text{ k}\Omega$ is to operate at 2.5 kHz . Select C for specified oscillator operation.	3	6

Q. 5	Solve Any Two of the following.		12
A)	Briefly explain about the IC 555 Block Diagram.	1	6
B)	Explain about adjustable three terminal positive and negative regulators.	1	6
C)	Write down the comparison of Linear power supply and SMPS.	5	6
	*** End ***		